METRIC

DOD-N-70329
16 August 1982

MILITARY SPECIFICATION

NUT, PLAIN, ROUND, BALL AND ROLLER BEARINGS, METRIC,
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nut, plain, round, ball and roller bearings, that is slotted on its periphery, for use in assemblies with ball and roller bearings.

2. APPLICABLE DOCUMENTS

2.1 Goverment documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

MILITARY

MIL-I-6866 - Inspection, Penetrant Method of
MIL-I-6868 - Inspection Process, Magnetic Particle

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5310

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DOD-P-16232 - Phosphate Coating, Heavy, Manganese or Zinc Base (For Ferrous Metals)
DOD-N-70329/1 - Nut, Plain, Round, Ball and Roller Bearings, Narrow Series, Metric

STANDARDS

FEDERAL

FED-STD-H28/21 - Screw-Thread Standards for Federal Services, Section 21, Metric Screw Threads
FED-STD-66 - Steel: Chemical Composition and Hardenability

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-1312 - Fasteners, Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM A380 - Cleaning and Descaling Stainless Steel Parts, Equipment, and Systems
ANSI B46.1 - Surface Texture

(Applications for copies should be addressed to the American National Standards Institute, Incorporated, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8 - Tension Testing of Metallic Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references other than specification sheets cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Nuts shall be made from the following materials, as specified (see 3.1).

3.2.1 Carbon steel. Nuts shall be made from carbon steel conforming to FED-STD-66, steel numbers 1005 to 1035, 1112 to 1115, 1117, 1118, 1211 to 1215. Carbon steel shall have minimum tensile strength of 345 MPa and a minimum yield strength of 172 MPa conforming to ASTM E8.

3.2.2 Corrosion-resistant steel. Nuts shall be made from corrosion-resistant steel conforming to FED-STD-66, steel numbers 302, 303, 304 or 316. Corrosion-resistant steel shall have a minimum tensile strength of 579 MPa and a minimum yield strength of 241 MPa conforming to ASTM E8.

3.2.3 Alloy steel. Nuts shall be made from alloy steel conforming to FED-STD-66, steel numbers 4130 to 4142, 8740 or 8742. Alloy steel shall have a minimum tensile strength of 758 MPa and a minimum yield strength of 676 MPa conforming to ASTM E8.

3.3 Design and construction.

3.3.1 Dimensions. Dimensions and tolerances shall conform to the dimensional requirements of the applicable specification sheet and shall apply after protective finishes.

3.3.2 Hardness. Nuts shall have the following hardness:

3.3.2.1 Carbon steel. The hardness of carbon steel shall be 55-92HRB.

3.3.2.2 Corrosion-resistant steel. Corrosion-resistant steel nuts shall be stress relief annealed and at a hardness of 79-90HRB.

3.3.2.3 Alloy steel. Alloy steel nuts shall be hardened by quenching in oil from the hardening temperature and tempered to a hardness of 24-32HRC.

3.3.3 Threads. Threads shall conform to the "M" profile and tolerance class 5H, in accordance with FED-STD-H28/21.

3.3.4 Tensile strength. Nuts shall have the following tensile strength.

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3.3.4.1 Carbon steel. Carbon steel nuts shall have a minimum tensile strength and minimum yield strength of 3.2.1.

3.3.4.2 Corrosion-resistant steel. Corrosion-resistant steel nuts shall have a minimum tensile strength and minimum yield strength of 3.2.2.

3.3.4.3 Alloy steel. Alloy steel nuts shall have a minimum tensile strength and minimum yield strength of 3.2.3.

3.4 Surface roughness. Nuts shall have a maximum surface roughness of 3.2 micrometers (µm), determined in accordance with ANSI-B46.1.

3.5 Protective finish. Nuts shall be protected with one of the following finishes, as specified (see 3.1).

3.5.1 Zinc coating. Zinc coating shall conform to QQ-Z-325, type II, class 3 (5.1 µm thick).

3.5.2 Cadmium plating. Cadmium plating shall conform to QQ-P-416, type II, class 3 (5.1 µm thick).

3.5.3 Phosphate coating. Phosphate coating shall conform to DOD-P-16232, type Z, class 2.

3.5.4 Cleaning and descaling. Corrosion-resistant steel shall be cleaned and descaled in accordance with ASTM A380.

3.6 Cracks, seams, inclusions, and flaws. Nuts shall be free from cracks, seams, inclusions, and flaws (see 4.5.5.).

3.7 Sharp edges. Unless otherwise specified on the applicable specification sheet, sharp edges shall be rounded, chamfered or broken, 0.15 to 0.40 mm.

3.8 Identification marking. Nuts shall be marked in accordance with MIL-STD-130.

3.9 Workmanship. The workmanship shall be uniform in quality and appearance and free from hanging burrs and slivers, gouges, porosity, objectionable scale or any other defects which may adversely affect the nut's serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Material inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 Material inspection. Material inspection shall consist of certification supported by data verifying that the materials used in fabricating nuts are in accordance with the requirements of 3.2 prior to such fabrication.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with table I.

TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Group A		
Examination	1/	4.5.1
Group B		
Hardness	3.3.2	4.5.2
Surface roughness	3.4	4.5.3
Protective finish	3.5	4.5.4
Cracks, seams, inclusions, and flaws	3.6	4.5.5
Group C		
Tensile strength	3.3.4	4.5.6

1/ All requirements not covered by tests.

4.4.1 Inspection lot. An inspection lot shall consist of all nuts covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.4.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.3 Sampling for group A inspection. Sampling shall be in accordance with MIL-STD-105 inspection level II with an Acceptable Quality Level (AQL) of 1.5 percent defective for major defects. For minor defects, inspection level I with an AQL of 4.0 percent defective shall apply.

4.4.3.1 Classification of defects. Defects shall be classified as specified in table II.

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TABLE II. Classification of defects.

Category	Defect
Critical	None defined.
Major	
101	Runout and parallelism of faces not as specified.
102	Threads not as specified.
103	Face diameter not as specified.
Minor	
201	Thickness not as specified.
202	Outside diameter not as specified.
203	Protective finish, if applicable, missing.
204	Burrs or sharp edges.
205	Other dimensional characteristics not covered above.
206	Workmanship not as specified.
207	Marking missing, incorrect, insufficient, illegible or not permanent.
208	Packaging not as specified.

4.4.4 Sampling for group B inspection. Sampling shall be in accordance with MIL-STD-105 inspection level S-2 with an AQL of 1.5 percent defective and may be made on sample units which have been subjected to and have passed group A inspection.

4.4.4.1 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order if the lot is accepted and the sample units are still within specified requirements.

4.4.5 Sampling for group C inspection. Sampling shall be in accordance with MIL-STD-105 inspection level S-2 with an AQL of 2.5 percent defective.

4.4.5.1 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract.

4.4.6 Sampling for protective finish test. Sampling for protective finish test shall be in accordance with the applicable specifications referenced in 3.5.

4.4.7 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of PPP-H-1581.

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4.5 Methods of inspection.

4.5.1 Examination. Sample nuts shall be thoroughly examined to determine conformance with this specification and applicable specification sheets with respect to all requirements not covered by tests.

4.5.2 Hardness. Nuts shall be tested for hardness in accordance with MIL-STD-1312 test method 6.

4.5.3 Surface roughness. Surface roughness shall be tested in accordance with ANSI B46.1.

4.5.4 Protective finish test. Test for protective finish shall be in accordance with the applicable specification referenced in 3.5.

4.5.5 Cracks, seams, inclusions, and flaws. The presence of cracks, seams, inclusions, and flaws in nuts shall be determined by one of the following methods of inspection, unless visual inspection discloses discontinuities which would preclude the necessity for these inspection means.

4.5.5.1 Magnetic particle inspection. Magnetic particle inspection shall be performed on noncorrosion-resistant steel nuts in accordance with MIL-I-6868.

4.5.5.2 Penetrant inspection. Penetrant inspection shall be performed on corrosion-resistant steel nuts in accordance with MIL-I-6866.

4.5.6 Tensile strength. Tensile strength shall be tested in accordance with MIL-STD-1312, test method 8 (metric).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. These nuts are primarily intended for use with keyed washers in securing and locking ball and roller bearings in position against shaft shoulders.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.
b. Applicable specification sheet part number (see 3.1).
c. Level (degree) of protection, in accordance with PPP-H-1581, ordering data (see 5.1).

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Custodians:
Army - AR
Navy - OS
Air Force - 99

Review activities:
Army - EA
Air Force - 11
DLA - IS
NSA - NS

User activities:
Army - AT, ME
Navy - MC, SH

Agent:
DLA - IS

Preparing activity:
Army - AR

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DOCUMENT IDENTIFIER AND TITLE DOD-N-70329

NUT, PLAIN, ROUND, BALL AND ROLLER BEARINGS, METRIC, GENERAL SPECIFICATION FOR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   [ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

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